

- 2 A binary star consists of two stars A and B. The two stars may be considered to be isolated in space. The centres of the two stars are separated by a constant distance, as illustrated in Fig. 2.1.

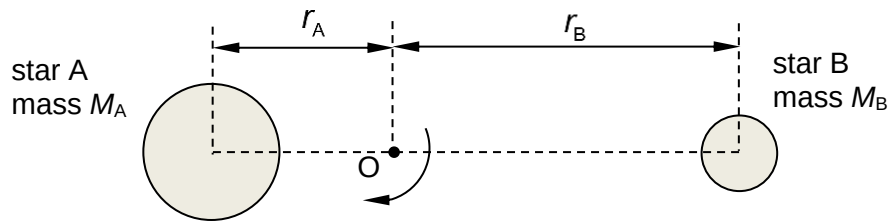


Fig. 2.1

Star A of mass M_A has a larger mass than star B of mass M_B such that $M_A = 4M_B$.

The stars are in circular orbits about each other such that the centre of their orbits is at a fixed point O. The radius of orbit of star A and star B are r_A and r_B respectively.

The period of each orbit is T .

	(a)	Explain why the two stars must always be directly opposite as they move in the circular orbit.	
		 [2]	
		Solution: The centripetal force acting on each star is provided by the gravitational force between the two stars. The direction of the gravitational force acts along the line joining their centres of mass (which is the diameter of the circular orbit).	1 1
	(b)	Show that $\frac{r_B}{r_A} = 4$. Explain your working.	
			[2]
		Solution: The gravitational force on each star provides the centripetal force required for its respective circular motion about O. Since the stars have the same period, they have the same angular velocity ω. For star A, $\frac{GM_A M_B}{(r_A + r_B)^2} = M_A r_A \omega^2$ (1) For star B, $\frac{GM_A M_B}{(r_A + r_B)^2} = M_B r_B \omega^2$ (2) ----- Equating (1) and (2) OR Since the stars have the same period, they have the same angular velocity. Therefore the magnitude of centripetal force acting on each star is equal. $M_A r_A \omega^2 = M_B r_B \omega^2$ (3)	1 1

		$\frac{r_B}{r_A} = \frac{M_A}{M_B} = \frac{4M_B}{M_B}$ $\frac{r_B}{r_A} = 4 \quad (\text{shown})$	
	(c)	If the period T is 104 days and the separation of the centres of the stars is 1.1×10^{11} m,	
	(i)	calculate the angular velocity of star A, and	
		angular velocity = rad s ⁻¹	[1]
	L1	Solution: $\text{angular velocity } \omega = \frac{2\pi}{T} = \frac{2\pi}{104 \times 24 \times 60 \times 60} = 7.0 \times 10^{-7} \text{ rad s}^{-1}$	1
	(ii)	determine the mass of each star.	
		$\text{mass } M_A \text{ of star A} = \dots\dots\dots \text{ kg}$ $\text{mass } M_B \text{ of star B} = \dots\dots\dots \text{ kg}$	[3]
	L2	Solution: $r_A + r_B = 1.1 \times 10^{11}$ $r_A + 4r_A = 1.1 \times 10^{11}$ $r_A = 2.2 \times 10^{10} \text{ m} \quad \text{and} \quad r_B = 8.8 \times 10^{10} \text{ m}$	1

The gravitational force on star A provides the centripetal force for its circular motion about O.

$$\frac{GM_A M_B}{(r_A + r_B)^2} = M_A r_A \omega^2$$

$$\frac{GM_B}{(r_A + r_B)^2} = r_A \omega^2 \Rightarrow M_B = \frac{\omega^2 r_A (r_A + r_B)^2}{G}$$

$$M_B = \frac{(7.0 \times 10^{-7})^2 (2.2 \times 10^{10})(1.1 \times 10^{11})^2}{(6.67 \times 10^{-11})}$$

$$M_B = 1.96 \times 10^{30} = 2.0 \times 10^{30} \text{ kg}$$

$$M_A = 7.8 \text{ or } 8.0 \times 10^{30} \text{ kg}$$

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